

Tactile Data Comics: Combining Step-by-step Presentation of Tactile Graphics with Verbal Narration for the Blind and Visually Impaired

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Abstract

Tactile graphics on a refreshable display have proven effective in enabling visually impaired people to comprehend pictorial content. To further evaluate the effectiveness of refreshable tactile displays in blind education, we designed *tactile data comics*, a method that combines step-by-step presentation of tactile graphics with verbal narration. We conducted a user study with sixteen visually impaired students to compare tactile data comics against verbal-only and static tactile graphics. Our findings show that tactile data comics significantly improve participants' comprehension and engagement during the learning experience. These empirical results suggest that the integration of refreshable tactile displays and tactile data comics represents a promising approach to enhancing the accessibility and effectiveness of tactile learning tools, with further exploration needed into affordability, usability, and extended applications.

CCS Concepts

• **Human-centered computing** → **Accessibility design and evaluation methods**.

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Keywords

Accessibility, Human-Computer Interaction, Refreshable tactile display

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1 Introduction

For centuries, tactile graphics have been used by individuals who are blind or have low vision (BLV) [38]. Traditionally, these graphics have been produced on paper using methods such as vacuum forming, braille embossing, or microcapsule paper [10, 46]. Among the tactile graphic media, refreshable tactile displays (RTDs) are unique in their ability to present dynamic content [46]. The working mechanism of RTDs is based on an array of thousands of small tactile pins, each capable of independent vertical movement to create a dynamic tactile surface. With the rise of maker technologies, various tactile display devices have emerged in the market or are currently in the prototype phase.

Past research studied how BLV individuals interact with static images (e.g., art, book diagrams, illustrations) [15, 25, 41, 43, 45] or dynamic content (e.g., sports, UML diagrams, animations) [18, 26, 31, 42] on RTDs. These applications demonstrate the potential of



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RTDs in enhancing the experience for BLV, particularly in interactive and dynamic learning contexts. However, evidence for the effectiveness of dynamic content presentation on RTDs in supporting blind students' learning remains limited. Despite positive user feedback, research lacks controlled studies measuring RTDs' impact on blind students' learning, relying instead on self-reported qualitative data [18]. Moreover, many existing studies predominantly involve older participants who are not students, leading to a gap in understanding how RTDs can effectively support blind students in educational settings [18, 51]. The lack of quantitative data in supporting (or refuting) the RTDs' pedagogical value and demographic discrepancy underscores the need for targeted research that includes blind students to accurately assess the educational potential of RTDs.

In this paper, we introduce a dynamic presentation method—*tactile data comics*—and conduct a controlled experiment to assess how RTDs and dynamic tactile graphics impact blind students' comprehension of educational materials. Tactile data comics reveal complex information in incremental, sequential components, each accompanied by a verbal narration to maintain cross-modal consistency for easier mental construction of the graphic. We conducted the experiment with 16 blind students using educational materials from four subjects—*mathematics, physics, maps, and literature*. The study included three conditions: *verbal-only*, *verbal with static graphics*, and *tactile data comics*. After interacting with the materials, participants answered quiz-like questions, providing objective data on the methods' effectiveness. At the end of the study, we also conducted semi-structured interviews to gather more detailed feedback on their user experience.

The results showed that tactile data comics significantly improved comprehension. The accuracy rates were 70.1% for the verbal condition, 70.1% for the static graphics condition, and 76.6% for the comic condition. Moreover, the findings indicate variation in participants' aptitude across different question types, with significant improvements in answer accuracy for the subjects of maps and literature. Participants also reported finding tactile data comics more engaging than both the verbal-only and static graphics conditions.

In summary, this paper contributes design and evaluation of tactile data comics, a procedural presentation method that integrates tactile graphics with verbal narratives to ensure cross-modal consistency. The demonstrated effectiveness of tactile data comics for some subjects, as well as qualitative findings, supports the value of RTDs in the educational context.

2 Related Work

We introduce related work for tactile display, roles of tactile illustrations in education, and interaction research around refreshable tactile displays.

2.1 Tactile Graphics and Tactile Displays

Tactile graphics are raised visual representations designed to convey information through touch for individuals who are blind or visually impaired, which play an essential role in their educational and professional activities [38]. Past work suggests that they are particularly useful for presenting graphics such as maps or charts

where spatial relationships are important [10]. By exercising exploration techniques with non-linear finger movement, one can touch and sense points, lines, or areas with slight elevation differences to understand the spatial relationships that the tactile pins/bumps represent [46]. Traditionally, tactile graphics have been accessed in printed form, produced using methods such as vacuum forming, braille embossing, or microcapsule paper [10, 46].



Figure 1: Existing refreshable tactile displays: (a) HyperBraille F, Metec.AG, 104x60 pins (40,000 Euro), (b) Graphiti Plus, Orbit Research, 60x40 pins (20,000 USD), (c) Dot Pad, Dot Incomp, 60x40 pins (10,000 USD)

An increasing variety of refreshable tactile displays (RTDs) are entering the market. Products currently available for consumers include HyperBraille F (Metec AG; 104x60 pin) [37], Graphiti Plus (Orbit Research, 60x40 pin) [53], and Dot Pad (Dot Incomp, 60x40 pin) [22] (Figure 1). These devices not only allow for content refreshing but can also dynamically adjust presentations in response to user interactions. These RTDs vary in cost [46]; more affordable models start at approximately 10,000 USD, while high-end devices could cost around 40,000 Euros. However, recent mechanical design innovations and technological advancements are making these devices increasingly affordable [59, 61], and we anticipate wider adoption of RTDs in the near future as a result. This increasing affordability and expected wider future adoption of the technology underscore the relevance of research on RTD interaction methods, including the past work from the HCI and ASSETS communities (e.g., [41], [18]), as well as our current work.

2.2 Tactile Graphics for Education and Beyond

Research has consistently demonstrated that integrating illustrations in educational materials enhances pedagogical outcomes [17, 34, 36]. In 1971, Paivio proposed the dual coding theory, which provides a theoretical foundation arguing for combining text and illustrations to facilitate learning, and has been supported by numerous subsequent studies [16, 28, 29]. For instance, Hibbing and Rankin-Erickson reported that students who struggle to create mental images while reading often experience comprehension difficulties; thus, providing visual aids could positively impact their educational outcomes [16]. Today, textbooks are filled with diverse visual illustrations, such as conceptual diagrams, data visualization, and photographs. These visual elements aid students in comprehending textual content, predicting story outcomes, and understanding how texts cohere around thematic points [29, 33, 35, 44]. In fact, some publishers limit text to no more than 30% of a page's space [44], emphasizing the essential role of visuals in textbooks. Conversely, this implies BLV students who cannot see the visual illustrations miss a significant portion of the educational content that is available for sighted peers. Our work explores how we could

use RTDs to enable BLV individuals access illustrative content for educational equity.

Tactile graphics are used in educational settings and beyond to support comprehension of textual content, like how visuals support comprehension of textual content [10]. Such assistive technologies are used to supplement child education [3, 4]. HCI research also aims to enhance how engineering subjects like circuit design are being taught [47, 48]. Applications of a combination of textual material and tactile graphics extend beyond education, including art [15, 50, 52], entertainment [26], and professional settings such as visual analytics [39] and UML [31]. Some researchers have even investigated the feasibility of real-time graphical information exchange between sighted and blind individuals [6]. Further research has also explored the use of assistive technologies for presenting dynamic visual content. For example, Holloway *et al.*'s study demonstrated the use of RTDs to present dynamic bird segments [18]. Given the importance of tactile graphics in educational settings, we investigate how our tactile display could be utilized through the evaluation of tactile data comics.

Tactile graphics have been extensively used for orientation and mobility (O&M) applications, like communicating relative positions of entities and routes between points of interest [9, 10, 49, 64]. While traditionally printed on paper, tactile maps have evolved with technological advancements. For instance, tactile displays have been employed for pre-travel route planning [23]. A series of works on 3D printed maps to aid orientation and mobility practices for blind individuals have been published in CHI and ASSETS [19–21]. Additionally, researchers have investigated methods to utilize RTDs for teaching spatial information in maps [30, 55, 63, 65]. These developments show ongoing advances in how technologies previously out of reach for the masses (e.g., 3D printing, refreshable tactile display) are evolving in how we present spatial information to blind people. Given tactile graphics' vital role in O&M applications, we also study how our tactile display could be used for map comprehension.

2.3 Blind Interaction with Graphical Contents

Tactual perception of a tactile graphic requires sequential exploration, as stimuli must be mentally combined to form a complete picture [46]. This process is unique among human senses, as the individual determines the sequence of perceived stimuli, resulting in a subjective perception that strongly depends on the exploration technique used [5]. This contrasts with visual perception, allowing a more direct comprehension of the overall image. Thus, in designing tactile data comics, we consider how we could enhance blind individuals' sequential process of consuming information by using a tactile graphic display.

Given the narrative nature of instructive material, we draw design inspiration from the emerging technique of data comics, which research has shown are effective for sighted people in consuming information as they break down complex, composite information into more digestible, sequential pieces [2]. Additionally, recent studies demonstrate that piece-by-piece narrative presentation of sonified and verbalized information [58] and bit-by-bit narrative communication [27] benefit blind users. These findings lead us to investigate whether such a sequential narrative technique can also effectively present coupled verbal narratives alongside instructional tactile

graphics, although no studies have yet confirmed the effectiveness of data comics—a visually complex and composite narrative form—in conveying information to blind individuals.

We note that related work, such as Lacey Lord's research, has explored translating traditional visual comics into audio-based formats, enabling blind readers to engage with comics through sound narration [32]. While we also use the term “comics”, our approach is more than simple transcoding. Our tactile data comics combine tactile elements with audio narration, allowing blind users to continuously perceive tactile information alongside auditory descriptions to build mental images gradually. Thus, our notion of “comics” extends beyond traditional formats by emphasizing a multimodal interaction that integrates touch and sound, offering a novel approach to non-visual storytelling.

3 Tactile Data Comics

In this section, we introduce the design of tactile data comics—a method that combines step-by-step tactile graphics with verbal narration.

3.1 The Concept of Tactile Data Comics

Tactile data comics use a step-by-step approach that breaks down complex images into smaller, digestible components. In this method, each component builds upon the previous one, retaining earlier content while introducing new elements (Figure 2). Additionally, each component is accompanied by verbal description of the presented content. The method ensures consistency between the two modalities, haptic and auditory, aiming to facilitate the user's comprehension of the presented material.

Tactile data comics separates itself from the existing presentation methods for educational tactile graphics in two ways. The first feature is a discrete frame-by-frame presentation of progressively evolving content, just like comics (Figure 2). This is in contrast to a continuous presentation, like animations, in [18, 62]. We made this design choice to reduce learning difficulties when perceiving the sequence of graphics displayed on the RTDs. Its progressive nature allows users to easily identify changes between frames; each component is built upon the previous one and retains the previously presented components until it transitions to the next image.

The second feature is pairing of each component with a verbal description, aiming to help users logically piece together the parts and gain a better understanding of the graphics being presented. The design of tactile data comics aims to extend the traditional learning experience, where verbal narration from teachers and static tactile images in textbooks were used to convey content. In the traditional method, regardless of which step of the content the teacher is introducing, the static tactile images from the textbooks remain unchanged and do not adapt to the teacher's narration. RTDs allow us to dynamically present a graphic that matches what is being explained. Our approach enables each step of the narration to correspond to a specific part of the tactile graphic, ensuring alignment between verbal and graphical information (Figure 2). For instance, in Fig. 2, the narration begins with “*This is a quarter circle...*” and is paired with a tactile graphic in step 1 that depicts the quarter circle. As the explanation progresses, additional elements are introduced in subsequent steps, building on the prior frame. This

approach ensures that blind students only interact with content directly related to the narration at each step. We believe this step-by-step, multi-modal presentation method significantly enhances information comprehension, making tactile data comics a valuable tool in blind education.

3.2 Tactile Data Comics Production

As this presentation technique is new, there is no established method for composing tactile data comics. For this paper, we compose tactile data comics contents by (i) manually segmenting static tactile images into step-by-step contents, and (ii) synchronizing the accompanying verbal narration to the segmented tactile contents. We break down the static tactile images based on the logical structure of the content in order to ensure the completeness of each component. Then, we create a verbal description to explain the content of each component. This is done manually by the research team members identifying the natural verbal segment in the original narration for each comic segment. When necessary, we edit the original narration for coherent and logical transition of the narration between segments, so that they can form a complete and coherent verbal story.

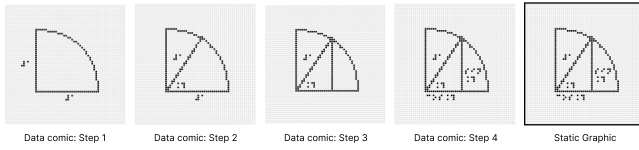


Figure 2: The four images on the left are tactile data comics of a geometric diagram explaining the Pythagorean theorem's principles, while the outlined image on the right is a static graphic of the same principles.

4 Method

To test tactile data comics' efficacy in improving blind users' comprehension of and engagement with educational material, we conducted a controlled user study with 16 blind students.

4.1 Refreshable Tactile Display

In this study, we utilized an internally developed tactile display to present tactile graphics (Figure 3). The device features a 15 cm by 15 cm display area and a 60×60 array of 3,600 refreshable pins (while commercially available tactile displays typically have 2,400–3,840 pins), spaced 2.5 mm apart, same as all the mainstream tactile displays on the market. It can refresh an image in less than 1 second, enabling the presentation of dynamic content. Thus, the device shares similar features with other commercially available tactile displays.

Our RTD exhibit two characteristics that are not common in traditional RTDs: mechanical actuation mechanism for pin extrusion and flip-latch structure for a fast refresh rate. First, in designing the actuators for the display pins, we moved away from piezoelectric ceramics—commonly used in both commercial tactile displays and our earlier prototype, due to their high material costs. Instead, we adopted mechanical actuation, which lowered the overall cost of the

display. Second, we incorporated a flip-latch structure to achieve a fast refresh rate while maintaining a simple actuation mechanism and reducing manufacturing costs. At its core, the design features a magnet-inserted eccentric wheel that rotates around a central axis (Figure 3.a). When activated by a solenoid, the wheel rotates, causing a braille pin to raise or lower. The eccentric design allows the wheel to climb up a protruding steel core as it flips, changing the pin's height. The flip-latch mechanism rotates between 0 and 180 degrees, switching the braille pins between “on” and “off” positions. Once the pin is latched in position, it maintains that state without requiring continuous current; it reduces power consumption and thus improves energy efficiency. Each pin contains a tiny iron ball, which ensures the pin remains magnetically attached to the eccentric wheel. This was an important design decision; without it, when the device was tilted, the pin could move due to gravity. Eight pins are combined to manufacture a braille cell (Figure 3.b). We created a matrix of 450 braille cell units to create a display (Figure 3.c). The display implemented with the flip-latch structure strikes the right balance of refresh time, energy consumption, and affordability. Importantly, (i) it supports a sufficiently fast refresh-rate to present dynamic contents to the users, and (ii) we expect the more affordable RTD design to contribute to the wider adoption of RTDs in the future.

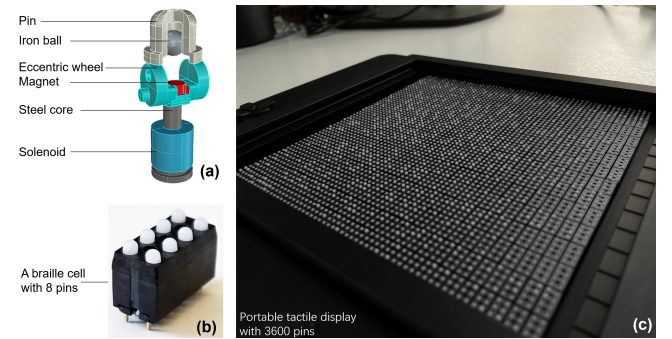


Figure 3: Hardware structure of current portable design. (a) flip latch actuator; (b) a group of 8 actuators; (c) 450 units of braille cells are combined to create a tactile display with 3600 pins.

4.2 Tactile Graphics

To investigate how different types of tactile graphics affect people's information comprehension, we prepared tactile graphics for four subjects commonly taught with tactile graphics in blind schools [10]: *mathematics*, *physics*, *maps*, and *literature*. We conducted in-depth discussions with teachers from blind schools to gather the teaching materials they currently use for each subject and to identify which materials would benefit from the integration of illustrations for enhanced learning. While the consultation helped guide content selection and ensured pedagogical alignment, the tactile data comics—including both graphics and audio narration—were created by the research team. Each tactile comic took approximately one hour to produce. We transformed existing illustrated materials from textbooks or created new ones, closely aligned with the curriculum,

into tactile data comics. For each subject, we created three materials. Thus, we had 12 sets of tactile comics overall (Table 1).

The tactile graphics ranged on a spectrum from representational to decorative [11, 17]. The tactile graphics for *mathematics*, *physics*, and *maps* were more representational; illustrations were designed to facilitate comprehension of accompanying textual material. The tactile graphics for the *literature* subject were more decorative, but still included representational information on important details about the accompanying story.

Like the example we presented above, the complete static tactile graphics were decomposed into multiple intermediate segments that gradually revealed the complete graphic. We split the graphics into segments based on the natural progression of their accompanying narratives. For each material for the *mathematics*, *physics*, and *maps* subjects, the static tactile graphic and the final segment of the data comic were equivalent. Because the passages for the *literature* subject were long, for each material, we had 1 to 4 static tactile graphics. Like other static graphics, we decomposed them into segments for tactile data comics.

All tactile graphics were designed to fit the dimensions of our tactile display (60×60 pins). Figure 4 illustrates examples of materials for *physics*, *maps*, and *literature*. Each material was paired with step-by-step a audible description, generated by a text-to-speech tool (<http://ttsqpt.cn>) to minimize variations in tone and reading speed between each experiment session. The full narrations and tactile comics for the 12 materials are available in the supplemental material.

4.3 Study Design

To evaluate the effectiveness of tactile data comics in conveying information and enhancing engagement for blind individuals, we conducted a within-subjects controlled study. The study tested three presentation methods: (1) verbal narration only (*verbal*); (2) verbal narration with static tactile graphics (*static*); and (3) tactile data comics (*comic*). The verbal and static conditions reflect how educational content is typically presented in classrooms, while the comic condition introduces a cross-modality, step-by-step approach designed to simplify complex concepts and graphics. We hypothesized that the comic condition would be the most effective in supporting comprehension.

Each participant engaged with 12 learning materials, using one presentation method per material. The order of presentation methods and learning materials was counterbalanced to prevent order effects. Participants explored the subjects in the fixed sequence of *mathematics*, *physics*, *maps*, and *literature*. During the study, the tactile display was connected to a smartphone via a Type-C cable for control. To ensure clarity, participants were notified whenever the graphic changed. To prevent variations in verbal narration from influencing the study results, we kept the audio content consistent across all three conditions. In the *verbal*-only and *static* conditions, the narration played alone. In the comics conditions, the narration synchronized with the tactile graphics. A 5-second pause was included at each section break within a material across all conditions.

After each material, participants were asked to answer six questions to assess their comprehension [33, 44]. This study's use of question-answering provides a more direct and accurate way to

verify the practical application of tactile data comics in blind education. We designed all the questions to be answerable solely through verbal narratives, without relying on tactile graphics. Half of the questions focused on recalling details from the narrations [35], while the other half were more conceptual and required participants' overall understanding of the material, evaluating not just memory but comprehension of the information as a whole [44].

At the end of the study, we conducted surveys and interviews to evaluate participants' perceptions of the different presentation methods. We used UEQ-S questions to assess each presentation method's perceived pragmatic and hedonic qualities [57]. UEQ-S had seven items for measuring the pleasure of use along with practical usability and utility in 7-point scale (e.g., '-3' indicating "very obstructive" and '+3' indicating "very supportive"). UEQ-S's design to assess the level of engagement and its brevity made it suitable for our study [56]. We designed interview questions to understand the participants' views on how tactile data comics differed from the other two methods and to gauge their interest in learning with our approach.

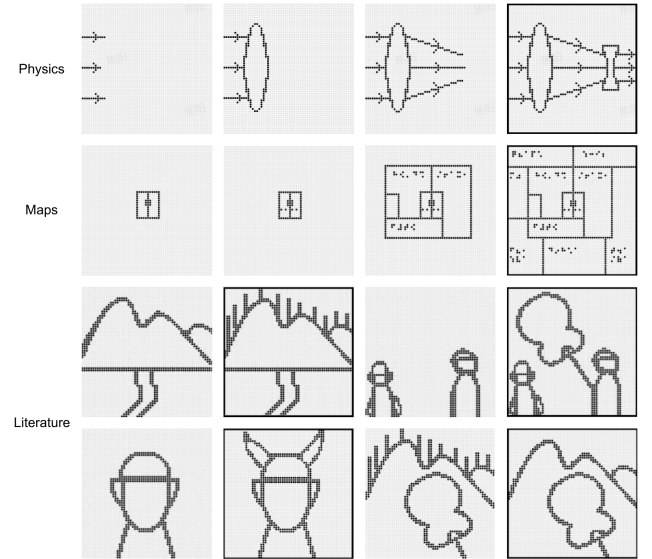


Figure 4: Examples of tactile data comics for *physics*, *maps*, and *literature* among the twelve educational materials that we used for the user study. A graphic with dark border represents the static tactile graphic that we used for each material. Because the narratives for the *literature* subject were longer, the accompanying tactile data comics were longer than those for other subjects, and we had more than one static graphics per material.

4.4 Participants

We recruited $N = 16$ participants from a blind college in the first author's city. The criteria for participant selection were that they must use braille. All participants were college students with prior exposure to paper-based tactile graphics, particularly in mathematics and physics. However, they had no prior experience with

Table 1: Themes for the twelve educational materials used in our user study. We had four subject categories: *mathematics*, *physics*, *maps*, and *literature*. For each subject, we prepared three materials.

Category	Material 1	Material 2	Material 3
Mathematics	Pythagorean Theorem	Trigonometric functions	Pythagorean identity
Physics	Reflection of light	Refraction of light	Principles of the telescope
Maps	District map	A suburban district map	A park map
Literature	“Journey to the West” Story 1	“Journey to the West” Story 2	“Journey to the West” Story 3

Table 2: Demographic data of our participants. All used braille and screen reader for learning. For the onset of visual impairments, early’ refers to acquiring blindness in early childhood at the age of six or younger, and late’ represents school age between 6 and 18 (the number in brackets represents the age of visual impairment onset).

Participant	Sex	Age	Onset of blindness
1	Male	24	Late (8)
2	Female	25	Late (13)
3	Female	19	Early
4	Female	26	Late (11)
5	Female	26	Early
6	Male	23	Early
7	Female	22	Early
8	Female	22	Late (8)
9	Female	24	Late (15)
10	Male	20	Late (18)
11	Male	24	Late (7)
12	Male	25	Late (12)
13	Male	27	Late (7)
14	Female	20	Early
15	Male	20	Early
16	Male	21	Early

refreshable tactile displays (RTDs) or dynamic tactile interfaces. Their demographic data is shown in Table 2.

4.5 Procedure

Each study session took place in a classroom at the school for blind students. A research team member led each session. Sessions were audio and video recorded using two smartphones: one placed in front to capture the entire experimenter-participant interaction and another positioned above the tactile display to capture hand interactions. The session involved four stages (Figure 5):

- (1) **Introduction:** After introducing ourselves and reviewing the study design with a participant, we obtained informed consent and collected demographic information. We explained that recordings would be used solely for academic research purposes and, upon consent, started audio and video recordings.
- (2) **Study Explanation and Practice:** The experimenter explained the task overview and learning materials. We demonstrated tactile display usage, and participants briefly experienced static and comic presentation methods.

- (3) **Engaging with Learning Materials:** We guided participants to engage with twelve learning materials using different presentation methods, following the exposure order described above. After each material, participants answered six questions, and the experimenter recorded the correctness of the response.
- (4) **Survey and Interview:** After engaging with all materials, the experimenter used UEQ-S to gather participants’ ratings of the three presentation methods. Following UEQ-S ratings, a semi-structured interview with five questions was conducted.

Each participant spent approximately 1 to 1.5 hours in completing the study session. We compensated the participants with about US\$20 for their participation.

5 Results

We report on four analyses: (i) Comprehension accuracy for three presentation methods; (ii) UEQ-S based analysis on subjective ranking of preferred presentation modes; (iii) Visual inspection of hand position visualization to understand how our participants explored the tactile graphics; (iv) Qualitatively investigating participants’ perception toward the presentation methods through thematic analysis.

5.1 Comprehension

Here, we examine how presentation methods and random effects affected comprehension of learning materials. Correctness for each question after engaging with a learning material was measured with a binary outcome (*i.e.*, correct or incorrect). Therefore, statistical tools like repeated measures ANOVA, which expects a continuous response, are not ideal. Thus, we used a generalized linear mixed model with a logit link function. The presentation method was the main effect. We treated materials’ subjects and participants as random effects.

5.1.1 Main Effect: Three Presentation Methods. Our participants answered the questions more accurately with the tactile data comics presentation. The accuracies were 70.1% for *verbal*, 70.1% for *static*, and 76.6% for *comic* (Figure 6). The difference in accuracy between *verbal* and *comic* conditions was statistically significant ($p = 0.039 < 0.05$), and so was the difference between the *static* and *comic* conditions ($p = 0.039 < 0.05$). We did not observe a statistically significant difference in accuracy between the verbal and static conditions. The result supports that tactile data comics help blind people comprehend learning materials.

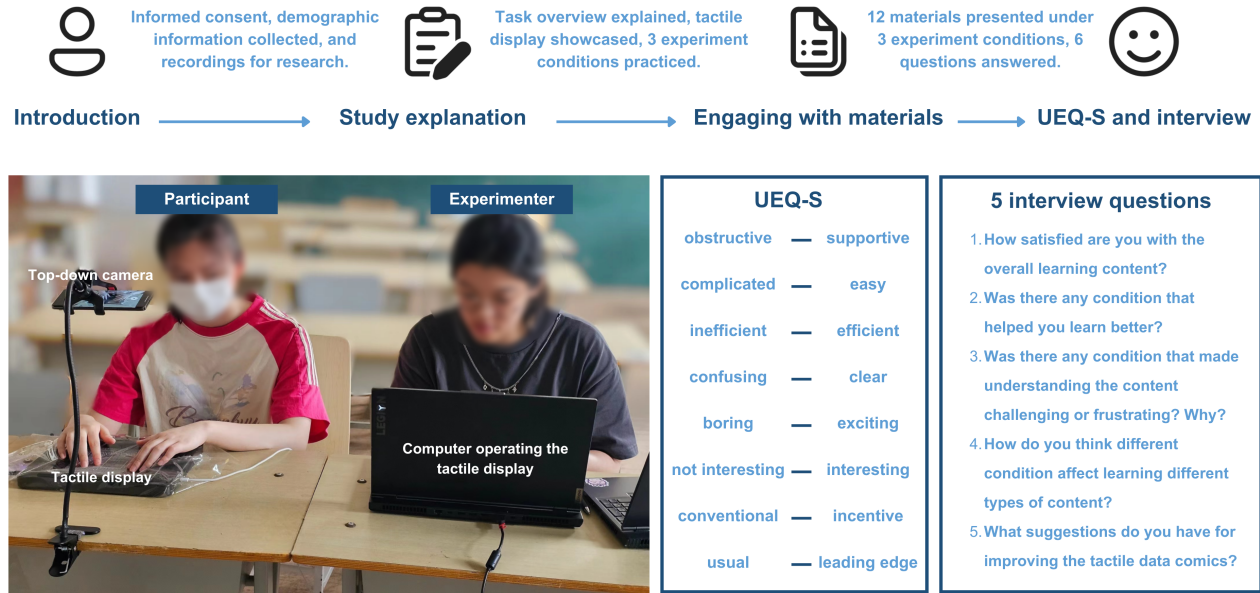


Figure 5: Overview of the study procedure, comprised of four stages: introduction; study explanation and practice; engaging with learning materials; and survey and interview. In the survey and interview stage, we used UEQ-S to assess subjective usability qualities, and five interview questions to gather our participants' perception toward presentation methods.

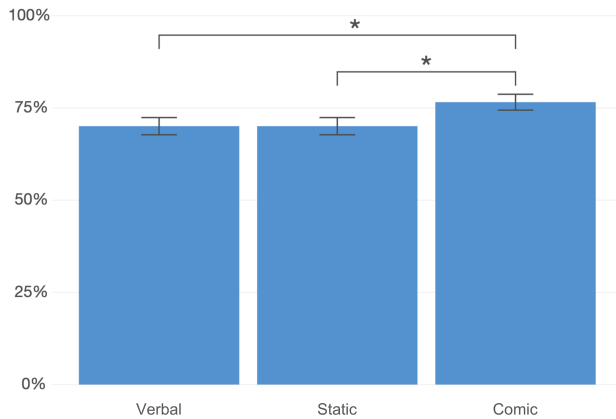


Figure 6: Comprehension accuracy for conditions: verbal, static, and comic. The accuracies are 70.1% for verbal narration (*verbal*), 70.1% for verbal narration with static tactile graphics (*static*), and 76.6% for verbal narration with tactile data comics (*comic*). The comic condition shows significantly higher accuracy compared to both verbal and static conditions.

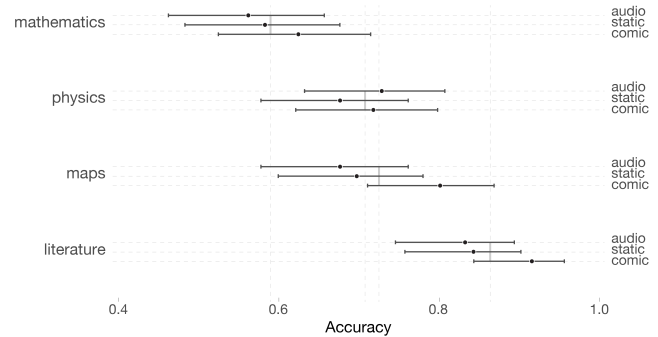


Figure 7: Comprehension accuracy for three conditions across four subjects. Each horizontal black segment shows the confidence interval of accuracy. Gray vertical bars represent per-subject mean accuracies (*math*: 0.59, *physics*: 0.71, *maps*: 0.73, *literature*: 0.87).

5.1.2 Subject Effect. Though we observed the significant benefit of tactile data comics for content comprehension, there seemed to be more nuance to the result. The questions' *subject* have influenced response accuracy. Overall, participants answered *literature* questions most accurately, with an average accuracy of 86.5%. For *map* and *physics* questions, accuracies were 72.6% and 70.8%, respectively. Participants' accuracy in answering *math* questions was lowest at

59.0% (Figure 7). This result indicates the variation in participants' aptitude for answering different question types in addition to the main effect of the presentation method.

We observed variations in presentation method effectiveness across subjects, though not statistically significant. Participants answered *map* and *literature* questions more accurately when exposed to the comic condition (Figure 7). On the other hand, we did not observe an interaction between subjects and conditions for *math* and *physics* subjects. The result may suggest that tactile data comics are suited to learning materials of particular subjects.

5.1.3 The Effect of Participants' Onset of Blindness. We grouped data by condition and blindness onset (early vs late) to explore possible differences in response accuracy. We used this classification to determine whether early-onset blindness and the resulting complete lack of exposure to visual educational content available to sighted peers influenced our comprehension results. Figure 8 suggests that the early onset group tended to have lower accuracy scores compared to the late group. However, this accuracy difference decreased in the comic condition. This may suggest that those who acquired blindness congenitally or early in life benefited more from tactile data comics. However, the overlapping confidence intervals suggest no significant differences and prevent us from making solid conclusions.

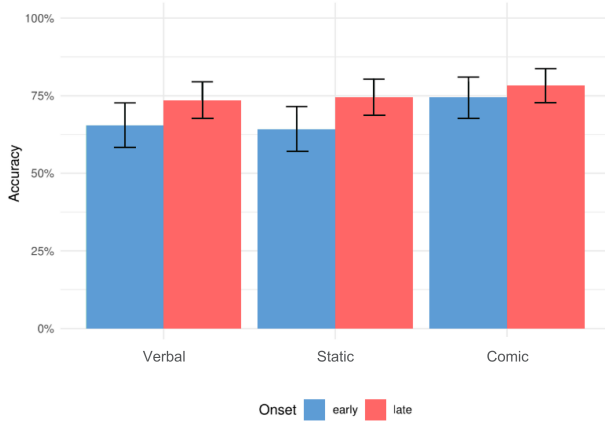


Figure 8: Comprehension accuracy by blindness onset (early vs late).

5.2 UEQ-S Responses

Perceived ease of interaction and engagement with the technology are important factors, particularly in a context like an educational setting, where such factors could impact retention. We collected subjective assessments of pragmatic and hedonic qualities for each presentation method. The stacked bar charts in Figure 9 correspond to distributions of the UEQ-S responses for each item. Positive sentiment, represented by '+1' to '+3', was most prevalent across all items for tactile data comics. The results show that participants felt learning with tactile data comics was easier and more engaging compared to learning with static tactile graphics or verbal narrations alone.

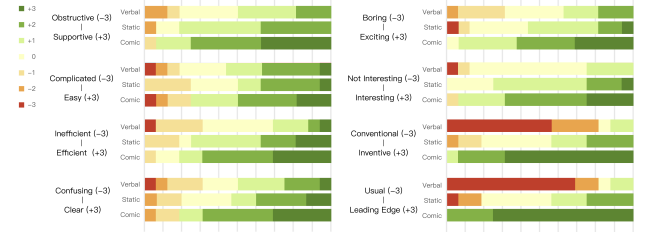


Figure 9: Distributions of UEQ scores across different interface conditions. Green to Yellow segments represent positive sentiment toward the usability of the presentation method, whereas Orange to Red segments represent negative. The **comic condition**' distributions predominantly filled with Green to Yellow segments show our participants' overall positive sentiments toward tactile data comics.

5.3 Hand Movement

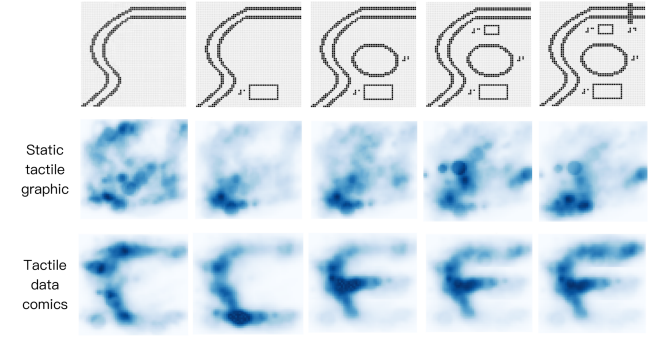


Figure 10: Distributions of where finger tips were placed on the tactile display while engaging with one of the *maps* learning material. The columns correspond to narrative segments. Each tactile graphic at the top correspond to a segment in this material's tactile data comic. In each heatmap, darker the blue, longer the participant kept their fingers at that position on the display. The heatmaps show that a participant moved his fingers all over the display across the segments with the *static* condition. In contrast, another participant with the *comic* condition had hand distributions that aligned more with what was displayed for the given segment.

The UEQ-S responses showed that the comic condition, where the content was presented step-by-step, was easier and clearer to interact with compared to the static condition, where we just presented the final tactile graphics. To better contextualize our participants' perceptions, we visualized their hand movements by tracking hands captured with the top-down camera. We used an off-the-shelf hand detection model (MediaPipe) and aligned the positions of the fingertips of both hands to the 60 × 60 pixel tactile display plane. We visualized the distribution of subjects' fingertips via a series of heatmaps, as shown in Figure 10.

To visualize the distribution of finger placement on the RTD, we followed the following algorithmic process (Figure 11). The process

begins by extracting hand gesture landmarks using a MediaPipe-based gesture recognition algorithm, which initially identifies points in the pixel coordinate system of the captured video footage. We then apply a coordinate transformation, mapping these coordinates onto our target 60×60 grid coordinate system. Given the specific characteristics of our video capture setup, we employ a non-distortion linear calibration method:

$$\lambda \begin{bmatrix} f_x \\ 1 \end{bmatrix} = \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} F_x \\ 1 \end{bmatrix} \quad (1)$$

where λ represents the scale transformation factor, R and T denote the rotation and translation matrices, respectively. For each fingertip F_x in the original pixel coordinate system, we apply this projection to obtain its corresponding grid coordinate f_x . This transformation allows us to track the spatial distribution of functionally significant fingertips—particularly the index, middle, and ring fingers of both hands, which primarily perform touching operations. The contribution of a specific fingertip f_x to the information correlation at a particular grid point x is quantified using the 2-Norm of their coordinate difference:

$$\rho(x, f_x) = \|x - f_x\|_2 \quad (2)$$

After that, we iterate over all grids to calculate the frame density of each grid, via a weighted sum of the negative exponentials of information correlation per fingertip:

$$\text{density}(x) = \sum_{f_x} w_{f_x} \cdot \exp(-\rho(x, f_x)) \quad (3)$$

where w_{f_x} is the corresponding weight of each fingertip regarding its contribution in touching performance. Eventually we can repeat this process every key frame to obtain the final heatmap in Fig. 11, which is capable of manifesting the general distribution of participants' touching gestures.

Figure 10 shows the fingertip position distributions of two participants, one in the static condition and one in the comic condition. Each column represents a comic segment (for the *static* condition, the narration corresponds to the segment, as the complete tactile graphic was presented from the beginning). Through the distribution pattern of the heatmaps in Figure 10, we find that compared to those using static graphics, participants interacting with our data comic better follow and understand the characteristics of the given images, showing higher similarity and faster convergence to the original point set, as clearly demonstrated by the heatmap series. Specifically, we observe that the heatmap series showcasing tactile data comics is visually more similar to the emerging patterns of a river, a farm, a park, a mall, and a small bridge over the river, while the series representing static graphics is much closer to a random distribution with fewer regional constraints. This suggests that subjects who interact with tactile data comics can better understand the pattern of the given image, rather than merely guessing or searching for the outline. This difference in hand movement may have led to the perceived differences in ease and clarity of interaction.

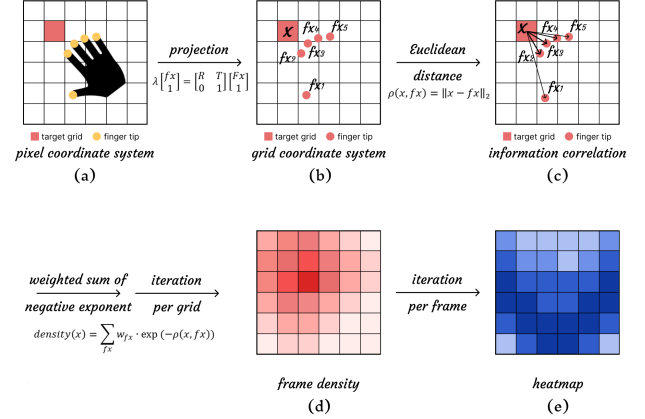


Figure 11: The process to obtain the target reading gesture heatmap. (a) A pixel coordinate system captured from the video footage. (b) The target 60×60 grid coordinate system after coordinate transformation. (c) Calculate the grid information correlation. (d) Calculate the frame density. (e) Iterate over all key frames to form a heatmap.

5.4 Interview Responses

We employed thematic analysis to identify key insights from the interview responses. We transcribed the semi-structured interview recordings using a verbal transcription tool. Two researchers independently conducted inductive and deductive coding using MaxQDA, focusing on participant actions relevant to our research goals. Any coding discrepancies were resolved through discussion, and an affinity diagram was used to group related codes and summarize emerging themes with internal connections.

5.4.1 Locate Information. Tactile data comics can help blind individuals more accurately locate information within a graphic by presenting content step by step and logically, aiding in understanding more complex and information-dense images.

With static tactile graphics, users sometimes find their touch positions misaligned with the verbal narration, leading to discrepancies in comprehension. In highly complex graphics, it becomes difficult for users to accurately discern the specific information referred to in the verbal narration. As P7 mentioned during the interview, “With static images, the content was too much, and I had to touch it repeatedly to differentiate the information.” Similarly, P10 noted, “For large-scale images in static formats, I couldn’t locate the information distribution, and when multiple patterns were overlapped, I couldn’t distinguish the boundaries, such as the blurred edges between squares and triangles in mathematical materials.”

These issues were effectively resolved and improved with tactile data comics. As tactile data comics ensure consistency between verbal narration and the tactile elements presented. For instance, P10 described, “Tactile data comics were quite satisfactory. The advantage of tactile data comics is that complex graphics can start with simple outlines and lines, gradually building an understanding, which makes it easier to absorb the information.” P12 also mentioned, “With

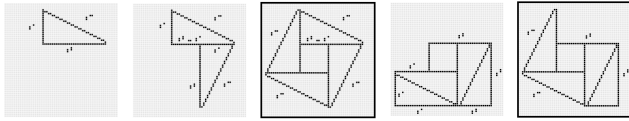


Figure 12: Tactile data comics of the Pythagorean Theorem from one of the 12 sets of experimental materials. The outline graphics are the static representations of the Pythagorean Theorem.

tactile data comics, starting from easy and gradually increasing in difficulty, step by step, this method greatly helped me understand the entire process more clearly.” Therefore, we conclude that tactile data comics are an excellent presentation method for images with complex information, making it easier for users to comprehend such information.

5.4.2 Comprehension of Abstract and Spatial Content. At the same time, the participants felt that using tactile data comics was highly effective in understanding abstract and spatial information.

Since blind individuals lack visual input, they cannot directly construct or understand spatial relationships, graphic outlines, and abstract concepts through visual cues, which means they often face more difficulties in grasping these aspects. For example, P13, discussing their experience with verbal narration, stated: “It’s very difficult to construct the corresponding graphic in my mind with purely verbal narration. Even if the verbal narration is clear, there can still be deviations when constructing the image, and once there’s a deviation, it becomes hard to understand the subsequent descriptions.”

In contrast, tactile data comics effectively compensate for the lack of visual information and enhance the user’s understanding of spatial and two-dimensional graphics. P8 pointed out: “I’m bad at math, but through this tactile learning method, especially in the material on the proof of the Pythagorean theorem, I could intuitively sense the changes in triangles through the changes in tactile data comics. I think that long-term learning this way would help build my spatial thinking in math.” As shown in Figure 12, the tactile data comics on the Pythagorean Theorem from the twelve experimental materials clearly present the transformation of triangles in a step-by-step manner. This approach reduces the discrepancies between blind users’ mental imagery and the corresponding audio descriptions. Unlike static graphics, which provide limited information, tactile data comics convey richer and more dynamic content, allowing users to better understand the entire process and the underlying spatial or abstract concepts. Therefore, this tactile comic method offers blind individuals a more intuitive and accessible way to grasp complex spatial and abstract information, enhancing their learning experience. This tactile data comic method helps blind individuals more intuitively grasp abstract and spatial information, significantly improving their learning experience.

5.4.3 Experience/Engagement. Tactile data comics not only enhance learning outcomes but also improve the overall learning experience by making it more interesting and engaging. Compared to verbal narration, our participants felt that tactile data comics make the learning material more vivid and immersive. P7 expressed: “I really liked the way tactile data comics presented the fourth type of

story. It felt like watching a movie or an animation-novel and interesting. It was my first time experiencing such a method. Before, when I watched accessible movies or animations, which relied solely on verbal descriptions, this tactile method of feeling while touching was very engaging.” The constantly changing graphic information also improves the user’s focus and attention during the learning process. Solely relying on auditory information for an extended period demands a high level of attention, making it difficult to maintain concentration over time. Tactile data comics, by constantly presenting new graphic details, keep users more engaged and focused. As P8 mentioned, “The tactile presentation method helps me maintain focus because the graphics are changing. Each change in the graphic brings new information, ensuring high levels of engagement.”

5.4.4 Disadvantages/Concerns over Tactile Data Comics. Although tactile data comics can improve learning outcomes, some participants reported that these graphics presented certain difficulties in information acquisition and comprehension. For students who had never previously encountered some genre of tactile graphics, they were more accustomed to relying solely on verbal narration. Years of training had equipped them with strong comprehension abilities based on auditory input alone. When tactile information was introduced, their ability to process information through both auditory and tactile channels within a limited time frame may have been affected. As P14 stated, “The form of tactile data comics felt unfamiliar to me. I think understanding tactile data comics requires cultivation from a young age, otherwise it’s hard to keep up with the verbal explanations.” P3 also mentioned encountering some problems in understanding tactile data comics, noting that their thinking process had not yet adapted to this method, but if trained from childhood, it could greatly enhance their ability to understand graphics. The feedback from some interviews related to this aspect differs from our original assumption, as we anticipated that the combination of two modalities might significantly enhance blind users’ learning. This suggests that further exploration of cross-modality presentation methods is necessary, taking into account the unique experiences and characteristics of blind education.

At the same time, some participants pointed out that tactile data comics increased the amount of information they needed to process. Even though tactile data comics and static images contain the same amount of content, each time the image refreshed, participants had to mentally reconstruct the framework of the graphic. For instance, P11 noted, “Tactile data comics require repeated understanding. It’s a lot of work, as every time the device refreshes a new image, I have to reconstruct the framework of the graphic in my mind.” This suggests that a key area of future research is designing presentation principles for tactile data comics that reduce the cognitive load on users.

5.4.5 Differences between Subjects. During the interviews, in addition to discussing their experiences with the three learning methods, users also shared their thoughts on learning across four different subjects. Six users believed that tactile data comics improved learning outcomes in mathematics and physics, as the dynamic presentation better aligned with the step-by-step processes in these subjects, helping the graphics correspond to each stage of the proofs and reducing the difficulty of understanding. As P4 stated, “Graphics are very helpful in subjects that require logic, like math.” For story-based

graphics, four users mentioned that the graphics did not significantly aid understanding, as “*text alone is sufficient to understand the story.*” However, five users noted that the graphics made the learning process more enjoyable and engaging. Regarding maps, users indicated that for simpler maps, static images were preferable, while tactile data comics were better suited for more information-dense maps, as they helped users precisely locate each landmark. “*The main advantage of tactile data comics was with the maps. There was a clear order, and each location was easy to find.*” (P10) Future research will focus on identifying the most effective presentation methods and design principles for different subjects, considering varying levels of information complexity, different cognitive processing styles, and determining which types of information are best suited for each presentation method.

6 Discussion

We designed tactile data comics, a step-by-step presentation of tactile graphics enabled by refreshable tactile displays. This method aims to improve blind users’ comprehension of educational materials. Our study with 16 blind participants showed significantly increased response accuracy when using tactile data comics, particularly for *map* content comprehension and *literature* narrative understanding. However, we observed no evident quantitative benefit for *math* and *physics* tasks, though some perceived their learning experience was better with tactile data comics compared to control conditions. The results highlight the potential of refreshable tactile displays’ dynamic nature in designing novel interaction methods for blind users. They also suggest variations in effectiveness across subjects, encouraging future exploration of novel interaction techniques on refreshable tactile displays. We divide the discussion into two parts: (i) the tactile data comics and (ii) the refreshable tactile display itself.

6.1 Tactile Data Comics

6.1.1 Long-Term Learning Effect. Participants expressed a desire to use tactile data comics for more learning materials across subjects, rather than being limited to the 12 materials we provided in the study. As one participant noted, “*I think that long-term learning this way would help build my spatial thinking in math.*” This reflects the perceived potential of tactile data comics in developing deeper cognitive skills, such as spatial reasoning, over time.

At the same time, some participants also pointed out challenges associated with this new learning method. For instance, P3 mentioned experiencing initial difficulties in understanding tactile data comics, noting that their thinking had not yet adapted to the combination of tactile and audio formats. They also expressed that if blind individuals were trained from an early age, their ability to comprehend multimodal information might be greatly improved. Therefore, further exploration is needed to understand how long-term use of refreshable tactile displays combined with tactile data comics might impact blind learners’ ability to comprehend information. This could offer important insights into how multimodal learning strategies can be systematically integrated into blind education.

6.1.2 Authoring Tactile Data Comics and Data Comic Principles. While TDCs offer a promising approach to improving learning outcomes, a significant challenge in increasing tactile display adoption and developing applications is content authoring. Creating TDCs typically involves more resources, including synchronization of audio with tactile frames. Our process of creating comic representation from existing materials was manual and time-consuming. Critical questions include: “*How can we translate visual content to tactile content more scalably?*” and “*How can we support authors in creating new tactile display contents easily?*”

While past work has explored authoring methods for static contents for tactile displays [6–8, 13, 48, 60], creating good tactile graphics remains challenging [46]. Given the rise of generative AI, an exciting next step is exploring whether image generation techniques can automatically create high-quality tactile content. Mukhiddinov *et al.* have initial work in this direction [40], but more research is needed to effectively generate good tactile graphics at scale. To reduce the manual effort traditionally involved in creating TDCs, we are currently investigating AI-based generation techniques. For STEM subjects such as mathematics and physics, algorithmic pipelines are being developed to automatically generate tactile graphics aligned with domain-specific rules and formulas. For text-heavy and interpretive domains like literature, large language models (LLMs) assist in generating narrative descriptions, which can then be mapped to tactile structures with minimal human refinement. These AI-supported approaches significantly reduce production time and open possibilities for scaling TDCs to broader educational applications. We aim to further advance this direction in future work, integrating automation with pedagogical insight to support long-term multimodal learning.

Additionally, while supporting easy and effective authoring of content for tactile data comics is one challenge, there are deeper questions around the design principles of TDCs. For example, the quotes from our participant indicated that comprehending tactile data comics induce more cognitive load than what we expected. Then, the question is, “*What is the upper bound of information density that many blind students can consume in interacting with tactile data comics?*” Unlike visual media, the tactile perception of the fingers is inherently limited to a small area [12], requiring careful management of layout complexity and graphical detail. Overly dense content can lead to cognitive overload or confusion. Thus, a key design consideration is determining which types of information are best suited to tactile data comics, and how much information can be effectively conveyed within each panel or sequence. Ultimately, making authoring more accessible and understanding the limits and affordances of tactile storytelling are both critical to scaling the use of tactile data comics in blind education.

6.2 Refreshable Tactile Displays

6.2.1 Function Improvement of Refreshable Tactile Displays. Based on our experimental observations and participant interview feedback, we found that the consistency between audio content and tactile graphics is crucial for blind users to accurately understand the presented images. Therefore, improving the refreshable tactile display by adding features such as hand movement sensing and audio instruction mode, where the device can detect which

part the user's hand is interacting with and provide synchronized audio narration, may enhance both students' comprehension and the overall usability of the device. As one participant suggested, adding auditory prompts to the machine's display would enable the tactile display to support the presentation of tactile data comics with cross-modal consistency (tactile and auditory), even without human assistance. Such enhancements could significantly expand the device's applicability and lower the barrier to independent use for blind individuals.

Moreover, our findings suggest that the sequential and narrative structure of tactile data comics may require users to revisit earlier steps to fully comprehend the content. To support this, future RTD designs could integrate features such as step indicators, progress markers, and navigational controls. For example, a "rewind" function would allow users to easily return to previous content, enabling self-paced exploration and reducing cognitive load. Collectively, these improvements would contribute to a more intuitive, adaptive, and educationally effective tactile interaction experience.

6.2.2 Affordability of Refreshable Tactile Displays. Our experimental results supported RTDs and tactile data comics can enrich educational experiences of blind individuals, enabling better comprehension of complex information. The result aligns with the past related work [18, 62]. However, the impact of tactile data comics (or other interaction methods on RTDs) hinges on whether RTD become more widely adopted. As we discussed earlier in the paper, currently available RTDs on the market remain expensive. Even relatively affordable models cost around 10,000 USD, which poses a significant barrier for access to RTDs. We emphasize the cost of RTDs because hardware availability directly determines the reach and scalability of educational technology. Regardless of how effective the interaction design or learning method may be, if schools, families, or educational institutions cannot afford the core hardware, these technological advancements will struggle to be implemented widely. Therefore, advancing RTD technology requires not only innovation in functionality and interaction but also careful consideration of cost structure and manufacturing feasibility to achieve widespread and lasting impact in educational practice.

This is precisely why we chose to develop our own tactile display, with a strong emphasis on cost-efficiency. Over more than ten years of iterative design and development, we have addressed several core challenges in creating a more affordable RTD, including the adoption of mechanical actuation and the flip-latch structure previously mentioned. We are continuing to iterate and enhance the device to further reduce its cost while expanding its functional capabilities, with the ultimate goal of making powerful tactile displays accessible to a broader blind community.

We are aiming to mass produce our device; we expect the per-unit retail price for our RTD to be approximately 7,000 USD for the first 10 units (which is already more affordable than many products on the market). This price reflects the costs of various components like pin headers, PCB board, and sensors, as well as the cost for software, manufacturing, and logistics. However, as production scales, these costs could decrease, and the profit margin could be reduced with increased sales, potentially leading to further improvements in affordability.

To estimate the effect of mass production to the device's future per-unit price, we use the Wright's model and cost simulation [1, 14]. The Wright's model, also known as a learning curve model, is a power law relationship that describes how the unit cost of a product decreases as the number of production increases. The model is widely used in estimating the effect of the economy of scale. It is expressed as follows: $Y = aX^b$. Here, a represents a constant of the first unit (*i.e.*, the unit cost without mass production) and b is a learning rate, which is a negative exponent indicating the rate at which the cost decreases as production increases. To perform simulation and estimate the effect of mass production on the future cost of our tactile display, we derive the value of a by fixing the price of the ten-unit price to 7,000 USD (actual per-unit price for our current product when produced in a bundle of 10 units). We then vary the learning rate b . Appropriate learning rates vary across industries and product types [24, 54]; here, we vary values from -0.05 to -0.2 . Figure 13 shows curves based on Wright's model over production quantity from 10 to 10,000 by ranging the learning rate $b = -0.05, -0.1, \text{ and } -0.2$.

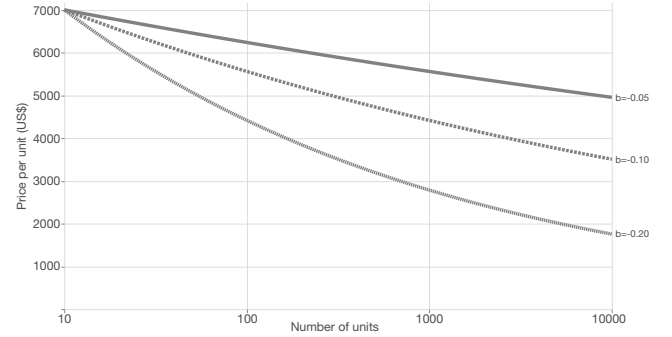


Figure 13: A projection of the unit price of our tactile display as a function of mass produced unit quantity, simulated using Wright's model. Each of the three curves represent a scenario with a different learning rate, b ($b = 0.05, 0.1, \text{ and } 0.2$). The curves share the intercept at US\$7,000—the planned retail price of our tactile display when produced as a bundle of ten units.

The figure suggests that the per-unit price could be halved at a learning rate $b = -0.1$ and production quantity $X = 10,000$ (approximately 3,500 USD). This is significantly less than US\$10,000. And with this price point, we believe it is not unrealistic to assume that RTDs will be adopted widely at many blind schools in the future.

6.3 Limitations

Beyond the directions for future exploration, several limitations of our study should be noted. First, we need to be cautious about whether the observed learning benefits can be generalized to age groups beyond those included in our study. Our participants were all college students, yet prior research has shown that children at different developmental stages have varying abilities to interpret tactile shapes and graphics [3]. Therefore, designs that worked

well for college students may not be equally effective for younger learners.

Second, we examined the effectiveness of tactile graphics, specifically static and comic-style representations, across a limited range of subjects and content. A more comprehensive understanding of how different types of tactile representations support learning would require broader studies across topics and domains.

Lastly, while our study focused on educational applications, as discussed earlier, refreshable tactile displays may have potential in other areas such as data accessibility, communication, or storytelling. However, our findings do not directly support or refute the effectiveness of interaction methods in these alternative contexts. Further research is needed to evaluate their applicability across domains.

7 Conclusion

In this work, we designed and evaluated tactile data comics, a method for presenting tactile graphics step by step, while maintaining consistency across auditory and haptic elements to take advantage of the refreshable nature of the tactile display. With a user study with 16 blind students, we showed that tactile data comics can facilitate comprehension of educational materials compared to control conditions (verbal-only presentation and static tactile graphic presentation). Qualitative data collected via UEQ-S questionnaire and semi-structured interview suggested that our participants were more engaged with the tactile data comic-based presentation than other conditions. Moving forward, we will continue to refine our self-developed refreshable tactile display to make it more affordable and accessible for the blind community. Additionally, we plan to expand our experiments to include a wider range of age groups, domains, and complexities to further assess the effectiveness of refreshable tactile displays and tactile data comics.

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